Commercial Performance & Unit Economics Intelligence

Executive Strategic Briefing: Margin Erosion, Discount Elasticity & Freight Audit

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Project Scope: End-to-End Commercial Analytics Case Study
Stack: Power BI (VertiPaq, Dynamic RLS)
Interactive Assets: [GitHub Repository]

Commercial Performance & Profitability Diagnostic

1. Current Performance:

- **Revenue: $1.92M** across **1,370** orders **(AOV: $1,410)**.
- **Net Profit: $182.3K** with an overall margin of **9.47%**.
- **Trajectory:** On track for **$3.8M** in annual sales, but bottom-line profit is heavily diluted by a few loss-making items.

2. Where We Lose Money:

- **Corporate Clients:** Margin dropped to just **2.62% ($17.2K** profit) because average discounts reached **5.06%**.
- **Unprofitable Categories:** 6 sub-categories generated **-$21.7K** in combined losses. The biggest issues are Tables **(-$7.2K overall)** and Corporate Office Machines **(-$16.7K)**.
- **South Region:** Margins collapsed to **-37.5%** in February due to expensive shipping on small orders.

3. Why It Happens:

- **Discounts Above 4.5% Hurt Profit:** Once discounts cross **4.5–5.0%**, orders quickly turn unprofitable without driving enough volume to make up for it.
- **Wrong Shipping for Heavy Goods:** Shipping bulky items like Tables via Regular Air causes a **-40.6%** loss.
- **Free Heavy Delivery:** We deliver heavy goods by truck with no extra freight fee, leaving only a **1.22%** margin on corporate deliveries.

4. Action Plan:

1. **Cap Corporate Discounts:** Set a hard **4.0%** discount limit on furniture and tech to keep deals profitable.
2. **Add Heavy Delivery Fee:** Introduce a mandatory fee **(e.g., $45)** for bulky freight like Tables and heavy office gear.
3. **Restrict Air Freight:** Stop shipping items over **15 kg** by Regular Air — move them to ground delivery or charge the customer for air express.

Ship Mode

Delivery Truck | Express Air | Regular Air

Customer Segment

Consumer | Corporate | Home Office | Small Business

1.92M	1.37K	1,410
Total Sales ($)	Total Orders	Avg Order Value ($)

Regional Sales & Profit Margin % Trend for All Products



Select row to filter trends

Product Category	Total Sales ($)	Total Profit ($)	Profit Margin %
Furniture	**660.70K**	**53.27K**	**8.06%**
Bookcases	107.80K	-0.93K	-0.86%
Chairs & Chairmats	261.07K	45.38K	17.38%
Office Furnishings	98.07K	16.07K	16.38%
Tables	193.76K	-7.24K	-3.74%
Office Supplies	**551.37K**	**64.13K**	**11.63%**
Appliances	82.20K	11.16K	13.58%
Binders and Binder Accessories	185.93K	54.76K	29.45%
Envelopes	10.48K	-1.22K	-11.59%
Labels	4.91K	0.44K	9.01%
Paper	55.81K	2.43K	4.36%
Pens & Art Supplies	26.07K	-4.93K	-18.91%
Rubber Bands	1.79K	-1.54K	-86.33%
Scissors, Rulers and Trimmers	6.75K	-3.28K	-48.54%
Storage & Organization	177.42K	6.31K	3.55%
Technology	**712.26K**	**64.90K**	**9.11%**
Computer Peripherals	96.26K	-2.53K	-2.63%
Copiers and Fax	99.07K	23.99K	24.22%
Office Machines	318.17K	7.36K	2.31%
Telephones and Communication	198.76K	36.08K	18.15%
Total	**1924.34K**	**182.31K**	**9.47%**

Ship Mode

Delivery Truck | Express Air | Regular Air

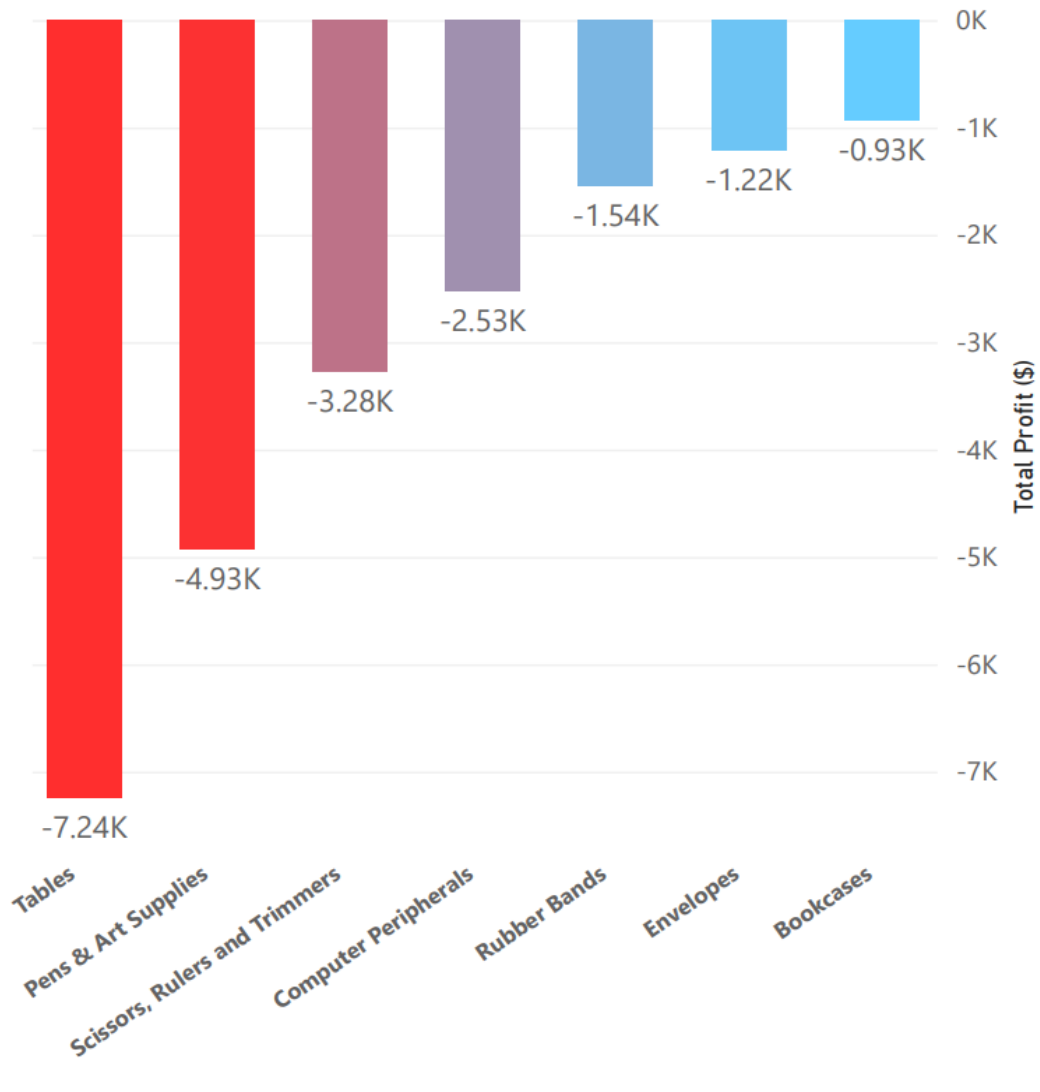
Customer Segment

Consumer | Corporate | Home Office | Small Business

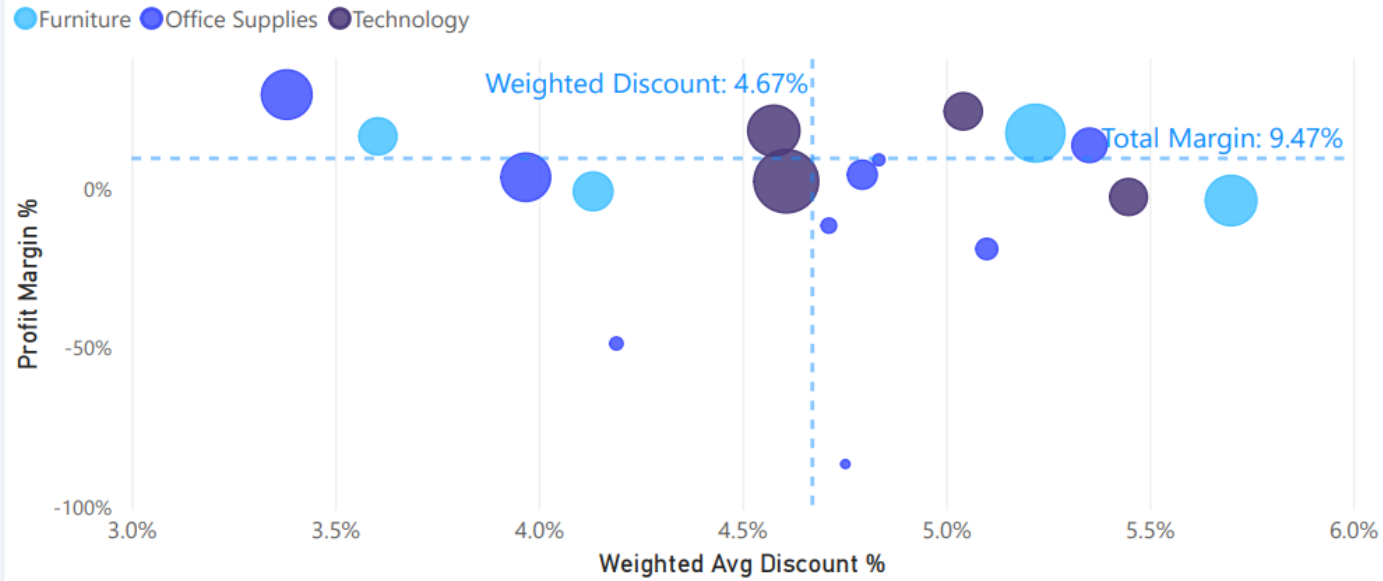
182.31K	9.47%	25.31K
Total Profit ($)	Profit Margin %	Total Shipping Cost ($)

Top Loss-Making Sub-Categories (Negative Profit)

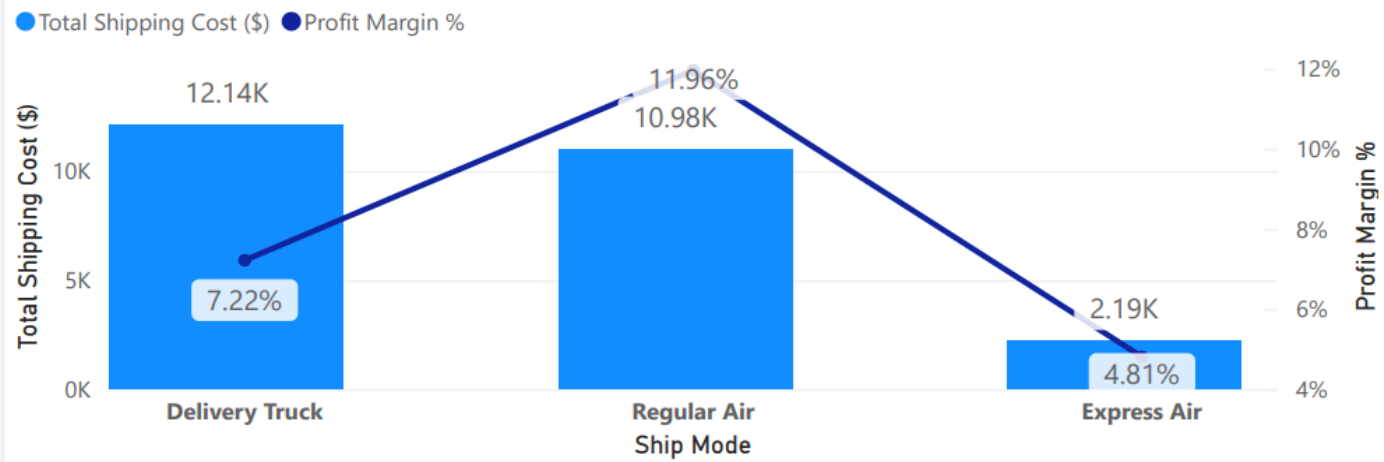
Total Loss for selected items: -$21.67K | Hover over bars for details

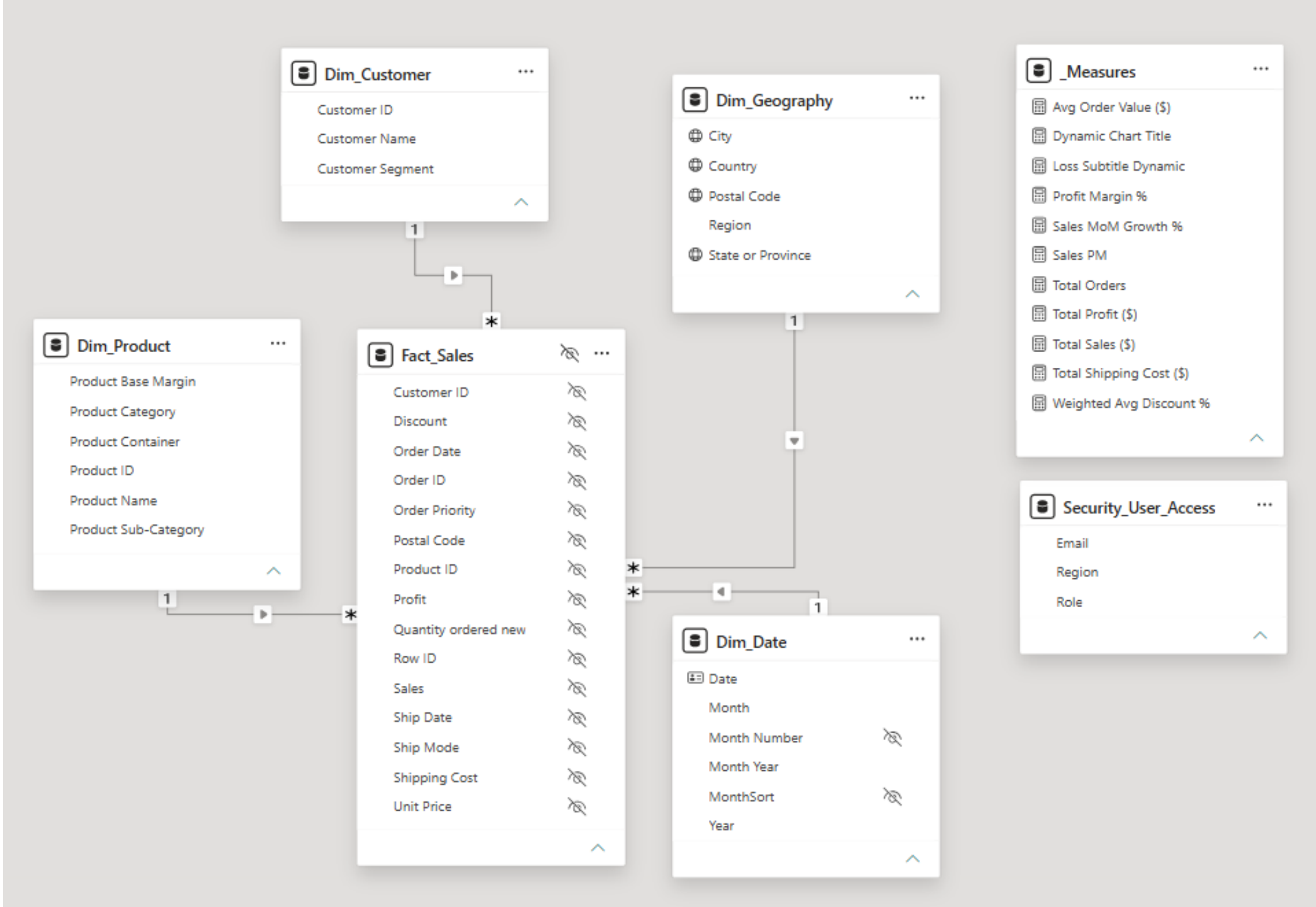


Profit Margin % vs. Weighted Discount % by Sub-Category



Total Shipping Cost ($) and Profit Margin % by Ship Mode




Dim_Customer
Customer ID
Customer Name
Customer Segment
Dim_Geography
City
Country
Postal Code
Region
State or Province
_Measures
Avg Order Value ($)
Dynamic Chart Title
Loss Subtitle Dynamic
Profit Margin %
Sales MoM Growth %
Sales PM
Total Orders
Total Profit ($)
Total Sales ($)
Total Shipping Cost ($)
Weighted Avg Discount %
Dim_Product
Product Base Margin
Product Category
Product Container
Product ID
Product Name
Product Sub-Category
Fact_Sales
Customer ID
Discount
Order Date
Order ID
Order Priority
Postal Code
Product ID
Profit
Quantity ordered new
Row ID
Sales
Ship Date
Ship Mode
Shipping Cost
Unit Price
Security_User_Access
Email
Region
Role
Dim_Date
Date
Month
Month Number
Month Year
MonthSort
Year
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